

Dynamic Rebalancing of Minimum-Variance Portfolios

LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Quantify portfolio drift.** Propagate pre-trade weights through asset returns and measure the resulting changes in risk, beta, and target tracking.
- **Design a rebalancing rule.** Distinguish calendar, threshold, and cost-aware optimization policies and state exactly when each policy trades.
- **Evaluate implementation effects.** Account for turnover, transaction costs, taxes, discrete shares, and information timing in a dynamic backtest.

1 An Optimal Portfolio Is a Time-Stamped Decision

A minimum-variance optimizer produces target weights for one information set, one covariance model, and one set of constraints. After prices move, a portfolio with fixed share counts no longer has those weights. Re-solving and trading continuously would chase every movement, consume liquidity, realize taxes, and expose the strategy to estimation noise. Dynamic rebalancing therefore asks a policy question: *when is the benefit of returning toward a target worth the cost and risk of trading?*

The distinction between a target and a policy is fundamental. A target $\mathbf{w}_t^*$ says what the portfolio would hold at decision time t if it could trade immediately under the current model. A rebalancing policy maps observable state—pre-trade weights, estimates, costs, tax lots, and constraints—to an action. The policy must be specified before evaluating later returns.

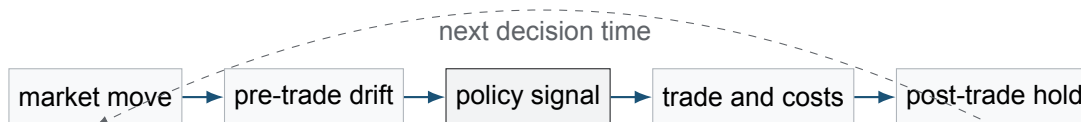


Figure 1: A rebalancing strategy is a repeated event sequence. Returns create pre-trade weights; the policy observes only information available at that time, decides whether to trade, pays implementation costs, and then holds the resulting portfolio.

2 How Buy-and-Hold Weights Drift

Let $\mathbf{w}_t^+ \in \mathbb{R}^m$ be post-trade risky-asset weights at time t , with $\mathbf{1}^\top \mathbf{w}_t^+ = 1$, and let $R_{i,t+1} > -1$ be asset i 's simple return from t to $t+1$. Before any new trade, the portfolio simple return is $R_{p,t+1} := (\mathbf{w}_t^+)^\top \mathbf{R}_{t+1}$ and the next pre-trade weights are

$$w_{i,t+1}^- = \frac{w_{i,t}^+(1 + R_{i,t+1})}{1 + R_{p,t+1}}, \quad i \in \{1, \dots, m\}. \quad (1)$$

The superscripts $+$ and $-$ mean immediately after and immediately before a trade, respectively. Equation (1) is a self-financing accounting identity before costs: asset values grow at different rates, while the denominator rescales them to fractions of total wealth.

With the course convention $r_{i,t+1} = (\Delta t)g_{i,t+1}$ for continuously compounded log return, $1 + R_{i,t+1} = \exp(r_{i,t+1})$. The same exact update is therefore

$$w_{i,t+1}^- = \frac{w_{i,t}^+ \exp((\Delta t)g_{i,t+1})}{\sum_j w_{j,t}^+ \exp((\Delta t)g_{j,t+1})}. \quad (2)$$

Do not replace the gross price relative by $1 + (\Delta t)g_{i,t+1}$ except as a stated small-move approximation.

Proposition 2.1. When weights do not drift

Assume $1 + R_{p,t+1} > 0$. For every asset with $w_{i,t}^+ > 0$, Eq. 1 satisfies $w_{i,t+1}^- = w_{i,t}^+$ if and only if

$$R_{i,t+1} = R_{p,t+1}. \quad (3)$$

Thus all positively weighted assets must earn the same return over the interval for every weight to remain unchanged. Assets held at zero weight impose no condition.

If a single-index model supplies beta vector β_t , the pre-trade portfolio beta becomes $\beta_{p,t+1}^- := (\mathbf{w}_{t+1}^-)^\top \beta_t$. Even if the beta estimates are held fixed, weight drift changes systematic exposure. With growth-rate covariance estimate $\Sigma_{g,t}$, modeled pre-trade variance is $(\mathbf{w}_{t+1}^-)^\top \Sigma_{g,t} \mathbf{w}_{t+1}^-$. A later model refit creates a second source of change; do not confuse parameter drift with mechanical weight drift.

COMPANION NOTEBOOK

L7b: Portfolio Drift After a Maximum-Sharpe Allocation →

Freeze initial share counts, propagate realized prices, reconstruct daily wealth fractions, and inspect how portfolio alpha and beta move away from their initial values.

3 Three Families of Rebalancing Rules

Let $\mathbf{w}_t^*$ be the current target and let $d(\mathbf{w}_t^-, \mathbf{w}_t^*)$ be a stated distance or risk-deviation measure. Common policies include:

1. **Calendar rebalancing.** Trade on predetermined dates—for example, monthly or quarterly—regardless of drift. It is simple to audit but can trade when nothing material changed and wait too long after a large move.
2. **Threshold rebalancing.** Trade when d exceeds a band, such as $\|\mathbf{w}_t^- - \mathbf{w}_t^*\|_\infty > \delta$ for tolerance $\delta > 0$. This creates a no-trade region but requires a convention for returning to the target, the nearest band edge, or somewhere between.
3. **Cost-aware optimization.** Choose the new post-trade weights directly by balancing modeled risk, target growth, and trading penalties. The action can be partial rather than all-or-nothing.

A hybrid policy checks thresholds only on scheduled dates. Every rule needs an explicit clock, input timestamp, and action. “Rebalance when needed” is not reproducible.

4 Turnover and the Cost of Returning to Target

Let $\Delta \mathbf{w}_t := \mathbf{w}_t^+ - \mathbf{w}_t^-$ be the planned weight change. For fully invested portfolios with no external cash flow, $\mathbf{1}^\top \Delta \mathbf{w}_t = 0$. The one-way turnover fraction is

$$\tau_t := \frac{1}{2} \|\Delta \mathbf{w}_t\|_1. \quad (4)$$

The factor $1/2$ avoids counting both sales and purchases as separate portfolio replacements. If a cost model charges a proportional rate $c_i \geq 0$ to every dollar bought or sold in asset i , the cost relative to pre-trade wealth is instead approximately $\sum_i c_i |\Delta w_{i,t}|$; state which convention is being reported.

One convex cost-aware update uses risk-aversion coefficient $\gamma > 0$ and turnover penalty $\lambda \geq 0$:

$$\begin{aligned} \min_{\mathbf{w} \in \mathbb{R}^m} \quad & \frac{\gamma}{2} \mathbf{w}^\top \Sigma_{g,t} \mathbf{w} - \bar{\mathbf{g}}_t^\top \mathbf{w} + \lambda \|\mathbf{w} - \mathbf{w}_t^-\|_1 \\ \text{subject to} \quad & \mathbf{1}^\top \mathbf{w} = 1, \quad \boldsymbol{\ell} \leq \mathbf{w} \leq \mathbf{u}. \end{aligned} \quad (5)$$

Here $\bar{\mathbf{g}}_t \in \mathbb{R}^m$ and $\Sigma_{g,t} \in \mathbb{R}^{m \times m}$ are information-time expected-growth and growth-rate covariance estimates, while $\boldsymbol{\ell}, \mathbf{u} \in \mathbb{R}^m$ are position bounds. If same-horizon log-return inputs are used instead, convert every input consistently: $\bar{\mathbf{r}}_t = (\Delta t) \bar{\mathbf{g}}_t$ and $\Sigma_{r,t} = (\Delta t)^2 \Sigma_{g,t}$. The penalty creates endogenous partial rebalancing: a trade occurs only when its modeled benefit overcomes its marginal penalty. Fixed commissions, bid–ask spreads, market impact, minimum lots, and taxes require richer and sometimes nonconvex models.

5 Backtesting a Dynamic Policy Without Looking Ahead

At each decision time, first mark the existing shares to current observable prices and compute $\mathbf{w}_t^-$. Then estimate model inputs using data whose timestamps precede the decision, construct $\mathbf{w}_t^*$ or solve Eq. 5, convert desired dollar changes into executable share quantities, apply fills and costs, and hold until the next permitted decision. Returns used to judge the action must occur strictly afterward.

Performance accounting should preserve the wealth recursion. If $C_t \geq 0$ is the dollar implementation cost paid at time t and W_t^- is pre-trade wealth, then post-trade investable wealth is

$$W_t^+ = W_t^- - C_t. \quad (6)$$

Ignoring Eq. 6 while changing weights grants the backtest free trading. Also record turnover, realized taxes or a declared pretax convention, constraint violations, cash residuals from integer shares, and failed or stale orders.

Use walk-forward evaluation: select policy parameters such as calendar frequency, threshold δ , and penalty λ on training or validation data, then freeze them for a later test period. Compare buy-and-hold, full-target rebalancing, and the proposed partial policy on identical price data and cost assumptions. The relevant question is not whether rebalancing preserves the model target perfectly; it is whether the net risk–return outcome improves after all implementation effects.

KEY TAKEAWAYS

In this lecture, we turned a static minimum-variance target into a time-indexed decision policy that observes drift, chooses an action, pays costs, and then holds.

- **Drift is mechanical.** Fixed share counts imply $w_{i,t+1}^- = w_{i,t}^+(1 + R_{i,t+1})/(1 + R_{p,t+1})$; unequal returns change weights and modeled exposures.
- **Rebalancing is a policy.** Calendar, threshold, and optimization rules differ in their decision clocks, triggers, and destinations, so each must be specified before testing.
- **Trading is part of the wealth equation.** Turnover, spreads, impact, taxes, and discrete shares can erase a model improvement and must be deducted at the time of action.

Next, begin the derivatives unit by separating an option's contractual payoff from the investor's premium-dependent profit.

ADDITIONAL READING

- André F. Perold and William F. Sharpe, “Dynamic Strategies for Asset Allocation,” *Financial Analysts Journal* 44(1), 16–27 (1988).
- Michael J. Magill and George M. Constantinides, “Portfolio Selection with Transactions Costs,” *Journal of Economic Theory* 13(2), 245–263 (1976).
- Richard Grinold and Ronald Kahn, *Active Portfolio Management*, 2nd ed. (1999), chapters on implementation, turnover, and transaction costs.

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